

SESSION 2 LAB EXPERIMENTS

1. Remove Duplicate Elements from an Array

```
#include <stdio.h>

int main()
{
    int n, arr[100], unique[100];

    int count = 0, found;

    printf("Enter size of array: ");
    scanf("%d", &n);

    printf("Enter array elements:\n");

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    for (int i = 0; i < n; i++)
    {
        found = 0;

        for (int j = 0; j < count; j++)
        {
            if (arr[i] == unique[j])
            {
                found = 1;
                break;
            }
        }

        if (found == 0)
            unique[count++] = arr[i];
    }
}
```

```

printf("Unique elements: ");
for (int i = 0; i < count; i++)
    printf("%d ", unique[i]);
return 0;
}

```

main.c	Run ▶	Output
<pre> 1 #include <stdio.h> 2 3 int main() 4 { 5 int n, arr[100], unique[100]; 6 int count = 0, found; 7 8 printf("Enter size of array: "); 9 scanf("%d", &n); 10 11 printf("Enter array elements:\n"); 12 for (int i = 0; i < n; i++) 13 scanf("%d", &arr[i]); 14 15 for (int i = 0; i < n; i++) 16 { 17 found = 0; 18 19 for (int j = 0; j < count; j++) </pre>		<pre> Enter size of array: 8 Enter array elements: 3 7 3 5 2 5 9 2 Unique elements: 3 7 5 2 9 </pre>

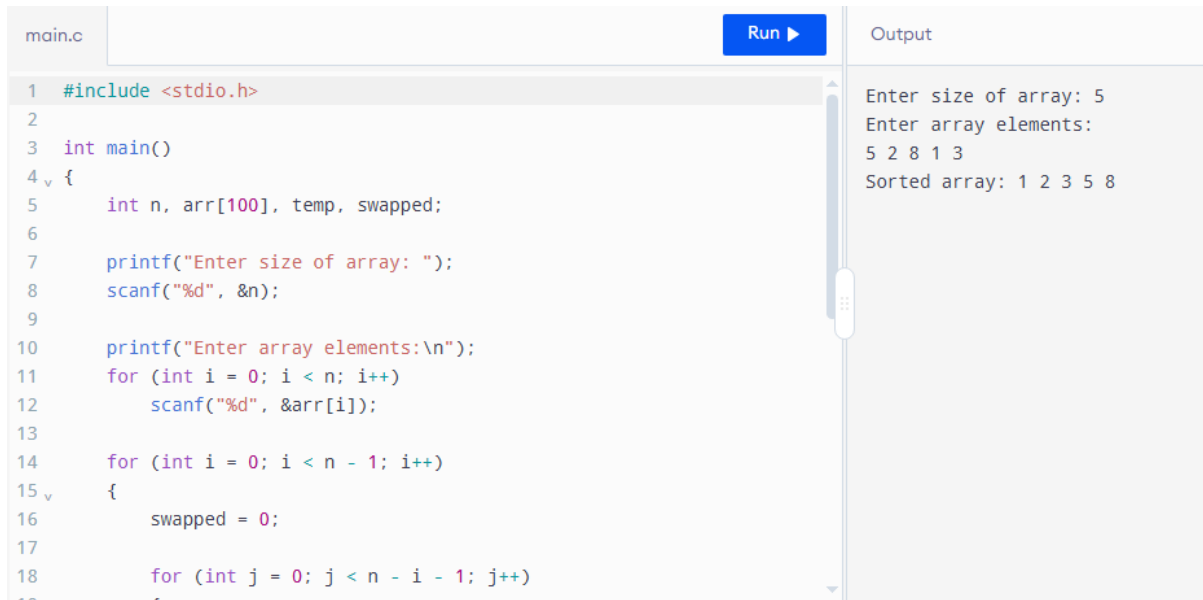
2. Bubble Sort

```
#include <stdio.h>

int main()
{
    int n, arr[100], temp, swapped;
    printf("Enter size of array: ");
    scanf("%d", &n);
    printf("Enter array elements:\n");
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    for (int i = 0; i < n - 1; i++)
    {
        swapped = 0;
        for (int j = 0; j < n - i - 1; j++)
        {
            if (arr[j] > arr[j + 1])
            {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;

                swapped = 1;
            }
        }
        if (swapped == 0)
            break;
    }
    printf("Sorted array: ");
```

```
    for (int i = 0; i < n; i++)  
        printf("%d ", arr[i]);  
    return 0;  
}
```



The image shows a code editor window with a file named 'main.c'. The code implements a sorting algorithm. To the right of the code editor is an 'Output' pane showing the program's execution results. A 'Run' button is located between the code editor and the output pane.

```
main.c Run ▶ Output  
1  #include <stdio.h>  
2  
3  int main()  
4  {  
5      int n, arr[100], temp, swapped;  
6  
7      printf("Enter size of array: ");  
8      scanf("%d", &n);  
9  
10     printf("Enter array elements:\n");  
11     for (int i = 0; i < n; i++)  
12         scanf("%d", &arr[i]);  
13  
14     for (int i = 0; i < n - 1; i++)  
15     {  
16         swapped = 0;  
17  
18         for (int j = 0; j < n - i - 1; j++)  
19             if (arr[j] > arr[j+1])  
20             {  
21                 temp = arr[j];  
22                 arr[j] = arr[j+1];  
23                 arr[j+1] = temp;  
24                 swapped = 1;  
25             }  
26     }  
27     if (swapped == 0)  
28         printf("Array is sorted.\n");  
29     else  
30         printf("Array is not sorted.\n");  
31     return 0;  
}
```

Enter size of array: 5
Enter array elements:
5 2 8 1 3
Sorted array: 1 2 3 5 8

3. Binary Search

```
#include <stdio.h>

int main()
{
    int n, arr[100], key;
    int low, high, mid, found = -1;

    printf("Enter size of sorted array: ");
    scanf("%d", &n);

    printf("Enter elements in ascending order:\n");
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Enter key to search: ");
    scanf("%d", &key);

    low = 0;
    high = n - 1;

    while (low <= high)
    {
        mid = low + (high - low) / 2;

        if (arr[mid] == key)
        {
            found = mid;
            break;
        }
    }
}
```

```

    }

    else if (arr[mid] < key)

        low = mid + 1;

    else

        high = mid - 1;

}

if (found != -1)

    printf("Element %d is found at position %d",

        key, found + 1);

else

    printf("Element %d is not found", key);

return 0;

}

```

main.c	Run ▶	Output
<pre> 1 #include <stdio.h> 2 3 int main() 4 { 5 int n, arr[100], key; 6 int low, high, mid, found = -1; 7 8 printf("Enter size of sorted array: "); 9 scanf("%d", &n); 10 11 printf("Enter elements in ascending order:\n"); 12 for (int i = 0; i < n; i++) 13 scanf("%d", &arr[i]); 14 15 printf("Enter key to search: "); 16 scanf("%d", &key); 17 18 low = 0; </pre>		<pre> Enter size of sorted array: 5 Enter elements in ascending order: 3 5 6 7 10 Enter key to search: 6 Element 6 is found at position 3 </pre>

4. Sort an Array in $O(n \log n)$ with Minimum Space

```
#include <stdio.h>
```

```
void swap(int *a, int *b)
```

```
{
```

```
    int temp = *a;
```

```
    *a = *b;
```

```
    *b = temp;
```

```
}
```

```
void heapify(int arr[], int n, int i)
```

```
{
```

```
    int largest = i;
```

```
    int left = 2 * i + 1;
```

```
    int right = 2 * i + 2;
```

```
    if (left < n && arr[left] > arr[largest])
```

```
        largest = left;
```

```
    if (right < n && arr[right] > arr[largest])
```

```
        largest = right;
```

```
    if (largest != i)
```

```
    {
```

```
        swap(&arr[i], &arr[largest]);
```

```
        heapify(arr, n, largest);
```

```
    }
```

```
}
```

```
void heapSort(int arr[], int n)
{
    for (int i = n / 2 - 1; i >= 0; i--)
        heapify(arr, n, i);

    for (int i = n - 1; i > 0; i--)
    {
        swap(&arr[0], &arr[i]);
        heapify(arr, i, 0);
    }
}
```

```
int main()
{
    int n, arr[100];

    printf("Enter size of array: ");
    scanf("%d", &n);

    printf("Enter array elements:\n");
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    heapSort(arr, n);

    printf("Sorted array: ");
    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);
```

return 0;

}

main.c	Run ▶	Output
<pre>1 #include <stdio.h> 2 3 void swap(int *a, int *b) 4 { 5 int temp = *a; 6 *a = *b; 7 *b = temp; 8 } 9 10 void heapify(int arr[], int n, int i) 11 { 12 int largest = i; 13 int left = 2 * i + 1; 14 int right = 2 * i + 2; 15 16 if (left < n && arr[left] > arr[largest]) 17 largest = left; 18</pre>		<pre>Enter size of array: 8 Enter array elements: 8 3 7 10 2 25 12 3 Sorted array: 2 3 3 7 8 10 12 25</pre>